

EXERCISE 6.8

Q.1 Find an equation of each of the following with respect to new parallel axes obtained by shifting the origin to the indicated point.

(i) $x^2 + 16y - 16 = 0$, $O'(0, 1)$

Solution:

Equations of transformation are

$$x = X + h \text{ and } y = Y + k$$

Here $h = 0, k = 1$, so $x = X$ and $y = Y + 1$

Put these values in given equation, we get

$$X^2 + 16(Y + 1) - 16 = 0$$

$$X^2 + 16Y + 16 - 16 = 0$$

$$\boxed{X^2 + 16Y = 0}$$

(ii) $4x^2 + y^2 + 16x - 10y + 37 = 0$, $O'(-2, 5)$

Solution:

Equations of transformation are

$$x = X + h \text{ and } y = Y + k$$

Here $h = -2, k = 5$, so $x = X - 2$ and $y = Y + 5$

Put these values in given equation, we get;

$$4(X - 2)^2 + (Y + 5)^2 + 16(X - 2) - 10(Y + 5) + 37 = 0$$

$$4(X^2 + 4 - 4X) + (Y^2 + 25 + 10Y) + 16X - 32 - 10Y - 50 + 37 = 0$$

$$4X^2 + 16 - 16X + Y^2 + 25 + 10Y + 16X - 32 - 10Y - 50 + 37 = 0$$

$$\boxed{4X^2 + Y^2 - 4 = 0}$$

(iii) $9x^2 + 4y^2 + 18x - 16y - 11 = 0$, $O'(-1, 2)$

Solution:

Equations of transformation are

$$x = X + h, y = Y + k$$

Here $h = -1, k = 2$, so $x = X - 1, y = Y + 2$

Put these values in given equation we get;

$$9(X - 1)^2 + 4(Y + 2)^2 + 18(X - 1) - 16(Y + 2) - 11 = 0$$

$$9(X^2 + 1 - 2X) + 4(Y^2 + 4 + 4Y) + 18(X - 1) - 16(Y + 2) - 11 = 0$$

$$9X^2 + 9 - 18X + 4Y^2 + 16 + 16Y + 18X - 18 - 16Y - 32 - 11 = 0$$

$$\boxed{9X^2 + 4Y^2 - 36 = 0}$$

(iv) $x^2 - y^2 + 4x + 8y - 11 = 0$, $O'(-2, 4)$

Solution:

Equations of transformation are

$$x = X + h \text{ and } y = Y + k$$

Here $h = -2, k = 4$, so $x = X - 2$ and $y = Y + 4$

Put these values in given equation, we get;

$$(X-2)^2 - (Y+4)^2 + 4(X-2) + 8(Y+4) - 11 = 0$$

$$X^2 + 4 - 4X - (Y^2 + 16 + 8Y) + 4(X-2) + 8(Y+4) - 11 = 0$$

$$X^2 + 4 - 4X - Y^2 - 16 - 8Y + 4X - 8 + 8Y + 32 - 11 = 0$$

$$\boxed{X^2 - Y^2 + 1 = 0}$$

(v) $9x^2 - 4y^2 + 36x + 8y - 4 = 0$, $O'(-2, 1)$

Solution:

Equations of transformation are

$$x = X + h \text{ and } y = Y + k$$

Here $h = -2, k = 1$, so $x = X - 2$ and $y = Y + 1$,

Put these values in given equation, we get;

$$9(X-2)^2 - 4(Y+1)^2 + 36(X-2) + 8(Y+1) - 4 = 0$$

$$9(X^2 + 4 - 4X) - 4(Y^2 + 2Y + 1) + 36(X-2) + 8(Y+1) - 4 = 0$$

$$9X^2 + 36 - 36X - 4Y^2 - 4 - 8Y + 36X - 72 + 8Y + 8 - 4 = 0$$

$$\boxed{9X^2 - 4Y^2 - 36 = 0}$$

Q.2 Find coordinates of the new origin (axes remaining parallel) so that first degree terms are removed from the transformed equation of each of the following. Also find the transformed equation.

(i) $3x^2 - 2y^2 + 24x + 12y + 24 = 0$

Solution:

$$3x^2 - 2y^2 + 24x + 12y + 24 = 0 \dots (i)$$

Let the coordinates of the new origin be (h, k) , so equations of transformation are

$$x = X + h, y = Y + k$$

Substituting these values of 'x' and 'y' in given equation we get;

$$3(X+h)^2 - 2(Y+k)^2 + 24(X+h) + 12(Y+k) + 24 = 0$$

$$3(X^2 + h^2 + 2Xh) - 2(Y^2 + k^2 + 2Yk) + 24X + 24h + 12Y + 12k + 24 = 0$$

$$3X^2 + 3h^2 + 6Xh - 2Y^2 - 2k^2 - 4kY + 24X + 24h + 12Y + 12k + 24 = 0$$

$$(3X^2 - Y^2) + X(6h + 24) + Y(-4k + 12) + 24h + 12k + 3h^2 - 2k^2 + 24 = 0 \dots (ii)$$

In order to remove first degree terms, we put coefficients of X and Y equal to zero

$$6h + 24 = 0 \Rightarrow h = -4$$

$$-4k + 12 = 0 \Rightarrow k = 3$$

Hence new origin is $O'(h, k) = O'(-4, 3)$

Now put value of h and k in (ii) we get;

$$3X^2 - 2Y^2 + X(6(-4) + 24) + Y(-4(3) + 12) + 24(-4) + 12(3) + 3(-4)^2 - 2(3)^2 + 24 = 0$$

$$3X^2 - 2Y^2 - 96 + 36 + 48 - 18 + 24 = 0$$

$$\boxed{3X^2 - 2Y^2 - 6 = 0}$$

(ii) $25x^2 + 9y^2 + 50x - 36y - 164 = 0$

Solution:

$$25x^2 + 9y^2 + 50x - 36y - 164 = 0 \dots (i)$$

Let the coordinates of the new origin be (h, k) , so equations of transformation are

$$x = X + h, y = Y + k$$

Substituting these values of 'x' and 'y' in given equation we get;

$$25(X + h)^2 + 9(Y + k)^2 + 50(X + h) - 36(Y + k) - 164 = 0$$

$$25(X^2 + h^2 + 2hX) + 9(Y^2 + k^2 + 2kY) + 50(X + h) - 36(Y + k) - 164 = 0$$

$$25X^2 + 9Y^2 + 50hX + 50X + 18kY - 36Y + 50h - 36k + 25h^2 + 9k^2 - 164 = 0$$

$$25X^2 + 9Y^2 + X(50h + 50) + Y(18k - 36) + 50h - 36k + 25h^2 + 9k^2 - 164 = 0 \dots (ii)$$

In order to remove first degree terms, we put coefficients of X and Y equal to zero

$$50h + 50 = 0 \Rightarrow h = -1$$

$$18k - 36 = 0 \Rightarrow k = 2$$

Hence new origin is $O'(h, k) = O'(-1, 2)$

Now put values of 'h' and 'k' in (ii) we get;

$$25X^2 + 9Y^2 + X(50(-1) + 50) + Y(18(2) - 36) + 50(-1) - 36(2) + 25(-1)^2 + 9(2)^2 - 164 = 0$$

$$\Rightarrow 25X^2 + 9Y^2 + X(0) + Y(0) - 50 - 72 + 25 + 36 - 164 = 0$$

$$\Rightarrow \boxed{25X^2 + 9Y^2 - 225 = 0}$$

(iii) $x^2 - y^2 - 6x + 2y + 7 = 0$

Solution:

$$x^2 - y^2 - 6x + 2y + 7 = 0 \dots (i)$$

Let the coordinates of the new origin be (h, k) , so equations of transformation are

$$x = X + h, y = Y + k$$

Substituting these values of 'x' and 'y' in given equation we get;

$$(X + h)^2 - (Y + k)^2 - 6(X + h) + 2(Y + k) + 7 = 0$$

$$X^2 + h^2 + 2Xh - Y^2 - k^2 - 2kY - 6X - 6h + 2Y + 2k + 7 = 0$$

$$X^2 - Y^2 + 2Xh - 6X - 2kY + 2Y - 6h + 2k + h^2 - k^2 + 7 = 0$$

$$X^2 - Y^2 + X(2h - 6) + Y(-2k + 2) - 6h + 2k + h^2 - k^2 + 7 = 0 \dots (ii)$$

In order to remove first degree terms, we put coefficients of x and y equal to zero,

$$2h - 6 = 0 \Rightarrow h = 3$$

$$-2k + 2 = 0 \Rightarrow k = 1$$

Hence new origin is $O'(h, k) = O'(3, 1)$

Now put values of h and k in (ii) we get;

$$X^2 - Y^2 + X(2(3) - 6) + Y(-2(1) + 2) - 6(3) + 2(1) + (3)^2 - (1)^2 + 7 = 0$$

$$X^2 - Y^2 + X(0) + Y(0) - 18 + 2 + 9 - 1 + 7 = 0$$

$$\boxed{X^2 - Y^2 - 1 = 0}$$

Q.3 In each of the following, find an equation referred to the new axes obtained by rotation of axes about the origin through the given angle:

(i) $xy = 1, \theta = 45^\circ$

Solution:

As we know that equations of transformations are

$$x = X \cos \theta - Y \sin \theta$$

$$y = X \sin \theta + Y \cos \theta$$

Put $\theta = 45^\circ$, we get

$$\begin{cases} x = X \cos 45^\circ - Y \sin 45^\circ \\ \text{and} \\ y = X \sin 45^\circ + Y \cos 45^\circ \end{cases}$$

$$\Rightarrow \begin{cases} x = \frac{X - Y}{\sqrt{2}} \\ \text{and} \\ y = \frac{X + Y}{\sqrt{2}} \end{cases}$$

Putting in given equation, we get

$$\left(\frac{X - Y}{\sqrt{2}} \right) \left(\frac{X + Y}{\sqrt{2}} \right) = 1$$

$$\frac{X^2 - Y^2}{2} = 1$$

$$\boxed{X^2 - Y^2 = 2} \text{ is the required}$$

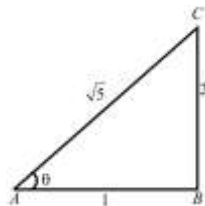
equation

(ii) $7x^2 - 8xy + y^2 - 9 = 0, \theta = \arctan 2$

Solution:

Here $\theta = \tan^{-1} 2 \Rightarrow \tan \theta = 2$

From right angle $\triangle ABC$



$$\sin \theta = \frac{2}{\sqrt{5}}, \cos \theta = \frac{1}{\sqrt{5}}$$

Using equations of transformations

$$x = X \cos \theta - Y \sin \theta$$

Put values of $\sin \theta$ and $\cos \theta$, we get

$$x = \frac{X}{\sqrt{5}} - \frac{2Y}{\sqrt{5}}$$

$$x = \frac{X - 2Y}{\sqrt{5}}$$

$$\text{As } y = X \sin \theta + Y \cos \theta$$

Put values of $\sin \theta$ and $\cos \theta$, we get

$$y = \frac{2X}{\sqrt{5}} + \frac{Y}{\sqrt{5}}$$

$$y = \frac{2X + Y}{\sqrt{5}}$$

Substituting the values of 'x' and 'y' in given equation, we get

$$\begin{aligned} 7 \left(\frac{X - 2Y}{\sqrt{5}} \right)^2 - 8 \left(\frac{X - 2Y}{\sqrt{5}} \right) \left(\frac{2X + Y}{\sqrt{5}} \right) + \left(\frac{2X + Y}{\sqrt{5}} \right)^2 - 9 = 0 \\ 7 \frac{(X^2 + 4Y^2 - 4XY)}{5} - 8 \frac{(2X^2 - 3XY - 2Y^2)}{5} + \frac{4X^2 + Y^2 + 4XY}{5} - 9 = 0 \end{aligned}$$

(Multiply by 5)

$$\begin{aligned} 7(X^2 + 4Y^2 - 4XY) - 8(2X^2 - 3XY - 2Y^2) + 4X^2 + Y^2 + 4XY - 45 = 0 \\ \Rightarrow (7 - 16 + 4)X^2 + (28 + 16 + 1)Y^2 \\ + (-28 + 24 + 4)XY - 45 = 0 \end{aligned}$$

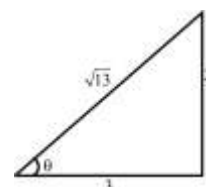
$$\Rightarrow -5X^2 + 45Y^2 - 45 = 0$$

$$\Rightarrow \boxed{X^2 - 9Y^2 + 9 = 0}$$

(iii) $9x^2 + 12xy + 4y^2 - x - y = 0, \theta = \tan^{-1} \left(\frac{2}{3} \right)$

Solution:

Here $\theta = \tan^{-1} \left(\frac{2}{3} \right) \Rightarrow \tan \theta = \frac{2}{3}$



From right triangle

$$\sin \theta = \frac{2}{\sqrt{13}}, \cos \theta = \frac{3}{\sqrt{13}}$$

Using equations of transformations

$$x = X \cos \theta - Y \sin \theta$$

$$y = X \sin \theta + Y \cos \theta$$

Using values of $\sin \theta$ and $\cos \theta$, we get

$$\begin{cases} x = \frac{3X - 2Y}{\sqrt{13}} \\ \text{and} \\ y = \frac{2X + 3Y}{\sqrt{13}} \end{cases}$$

Substituting these values of 'x' and 'y' in given equation, we get

$$\begin{aligned} 9 \left(\frac{3X - 2Y}{\sqrt{13}} \right)^2 + 12 \left(\frac{3X - 2Y}{\sqrt{13}} \right) \left(\frac{2X + 3Y}{\sqrt{13}} \right) + 4 \left(\frac{2X + 3Y}{\sqrt{13}} \right)^2 \\ - \left(\frac{3X - 2Y}{\sqrt{13}} \right) - \left(\frac{2X + 3Y}{\sqrt{13}} \right) = 0 \\ 9 \left(\frac{9X^2 + 4Y^2 - 12XY}{13} \right) + 12 \left(\frac{6X^2 + 5XY - 6Y^2}{13} \right) \\ + 4 \left(\frac{4X^2 + 9Y^2 + 12XY}{13} \right) \\ - \frac{3X}{\sqrt{13}} + \frac{2Y}{\sqrt{13}} - \frac{2X}{\sqrt{13}} - \frac{3Y}{\sqrt{13}} = 0 \end{aligned}$$

Multiply equation by 13, we get

$$\begin{aligned} \Rightarrow (81 + 72 + 16)X^2 + (36 - 72 + 36)Y^2 \\ + (-108 + 60 + 48)XY + (-3 - 2)\sqrt{13}X \\ + (2 - 3)\sqrt{13}Y = 0 \\ \Rightarrow 169X^2 - 5\sqrt{13}X - \sqrt{13}Y = 0 \end{aligned}$$

Dividing by $\sqrt{13}$ we get

$$\boxed{13\sqrt{13}X^2 - 5X - Y = 0}$$

(iv) $x^2 - 2xy + y^2 - 2\sqrt{2}x - 2\sqrt{2}y + 2 = 0, \theta = 45^\circ$

Solution:

Using equations of transformations

$$x = X \cos \theta - Y \sin \theta$$

$$y = X \sin \theta + Y \cos \theta$$

Put $\theta = 45^\circ$, we get

$$\begin{cases} x = X \cos 45^\circ - Y \sin 45^\circ \\ \text{and} \\ y = X \sin 45^\circ + Y \cos 45^\circ \end{cases}$$

$$\Rightarrow \begin{cases} x = \frac{X - Y}{\sqrt{2}} \\ \text{and} \\ y = \frac{X + Y}{\sqrt{2}} \end{cases}$$

Substituting these values of 'x' and

'y' in given equation, we get

$$\begin{aligned} \left(\frac{X - Y}{\sqrt{2}} \right)^2 - 2 \left(\frac{X - Y}{\sqrt{2}} \right) \left(\frac{X + Y}{\sqrt{2}} \right) + \left(\frac{X + Y}{\sqrt{2}} \right)^2 \\ - 2\sqrt{2} \left(\frac{X - Y}{\sqrt{2}} \right) - 2\sqrt{2} \left(\frac{X + Y}{\sqrt{2}} \right) + 2 = 0 \end{aligned}$$

$$\left(\frac{X - Y}{2} \right)^2 - \frac{2(X^2 - Y^2)}{2} + \frac{(X + Y)^2}{2} - 2(X - Y)$$

$$- 2(X + Y) + 2 = 0$$

Multiply the equation by 2 we get;

$$\begin{aligned} (X - Y)^2 - 2(X^2 - Y^2) + (X + Y)^2 \\ - 4(X - Y) - 4(X + Y) + 4 = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow (X + Y)^2 + (X - Y)^2 - 2(X^2 - Y^2) \\ - 4X + 4Y - 4X - 4Y + 4 = 0 \end{aligned}$$

$$\Rightarrow 2(X^2 + Y^2) - 2(X^2 - Y^2) - 8X + 4 = 0$$

$$2X^2 + 2Y^2 - 2X^2 + 2Y^2 - 8X + 4 = 0$$

$$4Y^2 - 8X + 4 = 0$$

$$\boxed{Y^2 - 2X + 1 = 0}$$

Q.4 Find measure of the angle through which the axes be rotated so that the product term XY is removed from the transformed equation. Also find the transformed equation.

(i) $2x^2 + 6xy + 10y^2 - 11 = 0$

Solution:

$$2x^2 + 6xy + 10y^2 - 11 = 0 \dots (i)$$

Comparing (i) with

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0,$$

we get

$$a = 2, b = 10, 2h = 6$$

As we know that angle of rotation is given by

$$\tan 2\theta = \frac{2h}{a-b}$$

$$\tan 2\theta = \frac{6}{2-10}$$

$$\tan 2\theta = \frac{6}{-8}$$

$$\tan 2\theta = -\frac{3}{4}$$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = -\frac{3}{4}$$

$$8 \tan \theta = -3 + 3 \tan^2 \theta$$

$$3 \tan^2 \theta - 8 \tan \theta - 3 = 0$$

$$3 \tan^2 \theta - 9 \tan \theta + \tan \theta - 3 = 0$$

$$3 \tan \theta (\tan \theta - 3) + 1(\tan \theta - 3) = 0$$

$$(\tan \theta - 3)(3 \tan \theta + 1) = 0$$

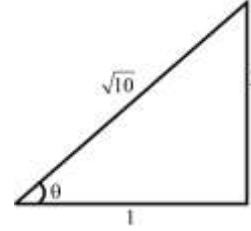
$$\tan \theta - 3 = 0, 3 \tan \theta + 1 = 0$$

$$\tan \theta = 3, \tan \theta = -\frac{1}{3}$$

$\therefore 0^\circ < \theta < 90^\circ$, so θ lies in 1st quadrant and in 1st quadrant $\tan \theta$ is positive, hence

$$\tan \theta = -\frac{1}{3} \text{ is neglected}$$

From $\tan \theta = 3$ we get;



$$\sin \theta = \frac{3}{\sqrt{10}}, \cos \theta = \frac{1}{\sqrt{10}}$$

Using equations of transformations

$$x = X \cos \theta - Y \sin \theta$$

$$y = X \sin \theta + Y \cos \theta$$

Using values of $\sin \theta$ and $\cos \theta$,

we get

$$\begin{cases} x = \frac{X - 3Y}{\sqrt{10}} \\ \text{and} \\ y = \frac{3X + Y}{\sqrt{10}} \end{cases}$$

Substituting values of 'x' and 'y' in (1), we get

$$\begin{aligned} 2 \left(\frac{X - 3Y}{\sqrt{10}} \right)^2 + 6 \left(\frac{X - 3Y}{\sqrt{10}} \right) \left(\frac{3X + Y}{\sqrt{10}} \right) \\ + 10 \left(\frac{3X + Y}{\sqrt{10}} \right)^2 - 11 = 0 \\ 2 \left(\frac{X^2 + 9Y^2 - 6XY}{10} \right) + \frac{(3X^2 - 8XY - 3Y^2)}{10} \\ + \frac{10(9X^2 + Y^2 + 6XY)}{10} - 11 = 0 \end{aligned}$$

Multiply the equation by 10, we get

$$\begin{aligned} 2(X^2 + 9Y^2 - 6XY) + (3X^2 - 8XY - 3Y^2) \\ + 10(9X^2 + Y^2 + 6XY) - 110 = 0 \\ \Rightarrow (2 + 18 + 90)X^2 + (18 - 18 + 10)Y^2 \\ + (-12 - 48 + 60)XY - 110 = 0 \end{aligned}$$

$$\Rightarrow 110X^2 + 10Y^2 - 110 = 0$$

Dividing by 10 on both sides

$$\boxed{11X^2 + Y^2 - 11 = 0}$$

(ii) $xy + 4x - 3y - 10 = 0$

Solution:

$$xy + 4x - 3y - 10 = 0 \dots (i)$$

Comparing (i) with

$$ax^2 + by^2 + 2gx + 2fy + 2hxy + c = 0,$$

we get

$$a = 0, b = 0, 2h = 1$$

As we know that angle of rotation is given by

$$\tan 2\theta = \frac{2h}{a-b}$$

$$\tan 2\theta = \frac{1}{0-0}$$

$$\tan 2\theta = \infty$$

$$2\theta = 90^\circ$$

$$\theta = 45^\circ$$

Using equations of transformations

$$x = X \cos \theta - Y \sin \theta$$

$$y = X \sin \theta + Y \cos \theta$$

Put $\theta = 45^\circ$, we get

$$\Rightarrow \begin{cases} x = X \cos 45^\circ - Y \sin 45^\circ \\ \text{and} \\ y = X \sin 45^\circ + Y \cos 45^\circ \end{cases}$$

$$\Rightarrow \begin{cases} x = \frac{X-Y}{\sqrt{2}} \\ \text{and} \\ y = \frac{X+Y}{\sqrt{2}} \end{cases}$$

Putting values of x and y in (1), we get

$$\left(\frac{X-Y}{\sqrt{2}}\right)\left(\frac{X+Y}{\sqrt{2}}\right) + 4\left(\frac{X-Y}{\sqrt{2}}\right) - 3\left(\frac{X+Y}{\sqrt{2}}\right) - 10 = 0$$

$$\frac{X^2 - Y^2}{2} + \frac{4(X-Y)}{\sqrt{2}} - \frac{3(X+Y)}{\sqrt{2}} - 10 = 0$$

Multiply the equation with 2 we get;

$$X^2 - Y^2 + 4\sqrt{2}(X-Y)$$

$$-3\sqrt{2}(X+Y) - 20 = 0$$

$$X^2 - Y^2 + 4\sqrt{2}X - 4\sqrt{2}Y$$

$$-3\sqrt{2}X - 3\sqrt{2}Y - 20 = 0$$

$$\boxed{X^2 - Y^2 + \sqrt{2}X - 7\sqrt{2}Y - 20 = 0}$$

(iii) $5x^2 - 6xy + 5y^2 - 8 = 0$

Solution:

$$5x^2 - 6xy + 5y^2 - 8 = 0 \dots (i)$$

Comparing (i) with

$$ax^2 + by^2 + 2gx + 2fy + 2hxy + c = 0,$$

we get

$$a = 5, b = 5, 2h = -6$$

As we know that angle of rotation is given by

$$\tan 2\theta = \frac{2h}{a-b}$$

$$\tan 2\theta = \frac{-6}{5-5}$$

$$\tan 2\theta = \frac{-6}{0} \Rightarrow \tan 2\theta = \infty$$

$$\Rightarrow 2\theta = 90^\circ \Rightarrow \theta = 45^\circ$$

Using equations of transformations

$$x = X \cos \theta - Y \sin \theta$$

$$y = X \sin \theta + Y \cos \theta$$

Put $\theta = 45^\circ$, we get

$$\begin{cases} x = X \cos 45^\circ - Y \sin 45^\circ \\ \text{and} \\ y = X \sin 45^\circ + Y \cos 45^\circ \end{cases}$$

$$\Rightarrow \begin{cases} x = \frac{X - Y}{\sqrt{2}} \\ \text{and} \\ y = \frac{X + Y}{\sqrt{2}} \end{cases}$$

Putting values of x and y in (i),

we get

$$5\left(\frac{X - Y}{\sqrt{2}}\right)^2 - 6\left(\frac{X - Y}{\sqrt{2}}\right)\left(\frac{X + Y}{\sqrt{2}}\right) + 5\left(\frac{X + Y}{\sqrt{2}}\right)^2 - 8 = 0$$

$$\frac{5(X - Y)^2}{2} - \frac{6(X^2 - Y^2)}{2} + \frac{5(X + Y)^2}{2} - 8 = 0$$

Multiply the equation by 2

$$5(X - Y)^2 - 6(X^2 - Y^2) + 5(X + Y)^2 - 16 = 0$$

$$5(X + Y)^2 + 5(X - Y)^2 - 6(X^2 - Y^2) - 16 = 0$$

$$5[2(X^2 + Y^2)] - 6(X^2 - Y^2) - 16 = 0$$

$$10X^2 + 10Y^2 - 6X^2 + 6Y^2 - 16 = 0$$

$$4X^2 + 16Y^2 - 16 = 0$$

$$\boxed{X^2 + 4Y^2 - 4 = 0}$$